Engineering - Undergraduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Calculate the length of a half-wave dipole antenna so as to optimallyradiate a signal of 300 MHz when the velocity factorfor the antenna element is (A) 0.70 (B) 1.0
   (a) 0.7 meters, 1 meter
   (b) 0.6 meters, 0.9 meters
   (c) 0.5 meters, 1 meter
   (d) 0.55 meters, 0.85 meters
   (e) 0.25 meters, 0.75 meters
2. The frequency range of a commercially broadcast FM signal is 88 to 108 MHz, with carrier swing of 125 kHz. Find the percentage modulation of the signal.
   (a) 110%
   (b) 125%
   (c) 50%
   (d) 62.5%
   (e) 75%
3. Findcurlaat the point (1, - 2, 1) fora= x^2y^2i+ 2xyzj+ z^2k(1)
   (a) (4, 4, 0)
   (b) (-2, 0, 0)
   (c) (0, 0, 4)
   (d) (4, 0, 0)
   (e) (2, 0, 0)
4. Which of the following windings are necessary in case of all dc machines?
   (a) star winding
   (b) delta winding
   (c) closed winding
   (d) parallel winding
   (e) zigzag winding
5. The current through a resistor of 2Ω is given by i(t) = cos^2. Find the energy dissipated in the resistor from t_0 = 0 to t_1 = 5 sec.
   (a) 5.75 J

(b) 7.50 J

(c) 4.75 J

(d) 1.25 J

(e) 3.75 J

True / False (10 marks)

Answer True or False

6. True or False: The lowest natural frequency for an automobile engine spring made of spring steel with the given parameters is 25.8 cycles/sec.

7. True or False: The equivalent resistance of five resistors in parallel with values of 10, 15, 20, 25, and 30 ohms is 4.12 ohms.

8. True or False: The information in the message about the dice sum of seven is exactly 1 bit, indicating a binary outcome.

9. True or False: A box that describes the effect of inputs on a control subsystem is known as a process box.

10. True or False: The coefficient of compressibility at constant temperature for an ideal gas is given by the expression (p^2) / (mRT).

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In the combustion of propane with 125% theoretical air, calculate the dew point and mole fractions of the products at 1 atm pressure. Discuss the implications of these values in combustion efficiency.

12. Derive the homogeneous solution for the voltage v_C(t) across a (1/2) F capacitor in a series RLC circuit with R = 4, L = 4H, i_L(0) = (1/4) A, and v_C(0) = 0. Discuss your findings.

End of Paper